Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

__show ip route ospf ______________________________ 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/1______ X D

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___64______ X D

Does R2 show up as an OSPF neighbor on R1? ___Yes_____ 2

Does R2 show up as an OSPF neighbor on R3? ___No_____ 2

What does this information tell you?

_____It tells that R2 was only advertising OSPF routes out of S0/0/0 initally, but not out of S0/0/1 ______ 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

__R2(config)#router ospf 1______________________________

__R2(config-router)#no passive-interface s0/0/1______________________________ 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __Serial0/0/1________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_the accumulated cost matric = 64+1 = 65, where 64 means R3 to R2 and 1 is for R2 G0/0 to network. 2

Is R2 listed as an OSPF neighbor to R3? ___Yes_____ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_Because the default reference bandwidth is only 100 Mbps, which makes every link that is 100 Mbps or faster have the same cost of 1 and cannot tell the difference. By increasing the reference bandwidth, OSPF can calculate more accurate costs ans actually prefer faster links over slower ones. 2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__For 192.168.3.0/24 the R1 S0/0/1 cost = 100000/128 = 781 with the cost of G0/0 which is 1, the total is equal to 782. ____ 2

_For 192.168.23.0/30: R1 S0/0/0 = 781 and R2 S0/0/1 still have the default bandwidth = 64. The total = 781+64 = 845

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

____Cost per serial link after the bandwidth set to 128 = 100000/128 = 781. As there are two sections from R1 to 192.168.23.0/24(R1 to R2 and R2 to R3), total = 781 +781 = 1562 ____________________ 2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___After manually set the cost on R1 S0/0/1 interface to ~~1565~~, this made the path though R3 more expensive than the path though R2. As OSPF always choose the path with the lowest cost, traffic goes through R2.____ 3

1566

____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   So it stays stable and does not keep changing every time when interface goes down. Using loopback addresses makes it predictable and way easier to troubleshoot since we actually know what ID to look for.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   Because we are only using point to point serial links between routers. DR/BDR elections only matter on multi-access networks like Ethernet where multiple routers are connected to the same switch. It is just direct connections in this lab so no election needed.

   4

3. Why would you want to set an OSPF interface to passive?

   So that it can stop wasting bandwidth and resources sending Hello packets out to the interfaces that don't have any OSPF neighbors.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   Areas. They break the network into smaller pieces so routers only need to know their own area and have summaries of the rest.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP routes because it has the lowest administrative distance which is 90.

   4